Quiz: Basic arithmetic

This quiz has 5 questions. The passing score is 4.

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Started on	Tuesday, January 14, 2025, 4:04 PM
State	Finished
Completed on	Tuesday, January 14, 2025, 4:11 PM
Time taken	6 mins 53 secs
Grade	**3.00** out of 5.00 (**60**%)

Question **1**

Correct

1.00 points out of 1.00

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The signed halfword 8AAA is, in decimal:

Select one:

- A. -30,038 ✔
- B. -5,888
- C. 23,210
- D. -20,000
- E. 5,101,010

Correct!

8AAA is in binary, **1**000 1010 1010 1010 (sign bit 1, so negative).

Converting 8AAA to positive:

Negative:	8AAA	1000 1010 1010 1010
Invert bits:	7555	0111 0101 0101 0101
Add 1:	7556	0111 0101 0101 0110

Convert hexadecimal 7556 to decimal:

$7 \times 16^3 + 5 \times 16^2 + 5 \times 16^1 + 6 \times 16^0 =$
$7 \times 4096 + 5 \times 256 + 5 \times 16 + 6 \times 1 =$
$28672 + 1280 + 80 + 6 = \mathbf{30,038}$

Halfword 7556 is decimal 30,038.

Halfword 8AAA is **decimal -30,038**.

The correct answer is: -30,038

Question **2**

Correct

1.00 points out of 1.00

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The decimal number 8,000 can be represented as (pick the two correct answers)

Select one or more:

- [x] A. Fullword 00001F40 ✔
- [ ] B. Fullword FFFFE0C0
- [ ] C. Fullword 0000E0C0
- [x] D. Halfword 1F40 ✔
- [ ] E. Halfword E0C0

Correct!

Converting decimal 8,000 to hexadecimal:

8000 / 16 = 500, with remainder **0**
500 / 16 = 31, with remainder **4**
31 / 16 = 1, with remainder 15 (hex **F**)
1 / 16 = 0, with remainder **1**

Decimal 8,000 is hexadecimal **1F40**. This is:

- Halfword **1F40**
- Fullword **00001F40**

The correct answers are: Halfword 1F40, Fullword 00001F40

Question **3**

Correct

1.00 points out of 1.00

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The decimal number 1234, converted to hexadecimal and placed in a fullword, is:

Select one:

- A. FFFFF4D2
- B. The result is too small for a fullword and must be placed in a halfword
- C. FFFFF4D0
- D. 000004D0
- E. 000004D2 ✔

Correct!

1234 / 16 = 77 with remainder **2**

77 / 16 = 4 with remainder **13 (hexadecimal D)**

4 / 16 = 0 with remainder **4**

Decimal 1234 is hexadecimal **4D2. In a fullword: 000004D2.**

The correct answer is: 000004D2

Question **4**

Incorrect

0.00 points out of 1.00

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The decimal number **-**32768 can be represented as (pick all that apply):

Select one or more:

- [ ] A. Fullword FFFF8000
- [x] B. Halfword 8000 ✔
- [x] C. Doubleword 0000000000008000 ✖
- [x] D. Fullword 00008000 ✖
- [ ] E. Doubleword FFFFFFFFFFFF8000

-32768 is, in a binary halfword, **8000**. Bit 0, the sign bit, is 1 (hexadecimal **8** is binary **1**000), indicating a negative number.

To represent the number as a **fullword**, we propagate the sign (**binary 1**) to the left: **FFFF8000**.

Same to represent the number as a **doubleword**: **FFFFFFFFFFFF8000**.

The correct answers are: Halfword 8000, Doubleword FFFFFFFFFFFF8000, Fullword FFFF8000

Question **5**

Incorrect

0.00 points out of 1.00

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(Assuming the numbers are fullword integers): **00000010 - 000F0010 = FFF10000**

Select one:

- True
- False ✖

000F0010 (**decimal 983,056**) is bigger than **00000010** (**decimal 16**), so we'll calculate **000F0010 - 00000010** and change the sign of the result.

```
  000F0010
- 00000010
--------------
  000F0000 decimal 983,040
```

To change the sign of **000F0000**, we invert the bits and add 1:

```
000F0000
FFF0FFFF  (bits inverted)
         +  1
  ------------
  FFF10000  result of 00000010 - 000F0010 (decimal -983,040)
```

The correct answer is 'True'.